AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

# Particulate Matter
## Monitoring & Health Effects

Issue 2026-10-23 | Entry window 23 October 2026 (full day) | 24 records screened in scope

**24** RECORDS IN SCOPE (124 EXCLUDED)

**8** OF 10 SUBTOPIC BANDS POPULATED

**5** EFFECT ESTIMATES WITH A CI

**5** TIER-A RECORDS

**3** RECORDS CARRIED ON METADATA ALONE

## Signal of the day

### 96 paired comparisons: optical dust monitors read 36 % above gravimetry, and site-specific calibration cuts that to 13 %.

A PRISMA meta-analysis of 112 occupational studies pooled 96 real-time-monitor vs gravimetric pairs (DustTrak, SidePak, pDR) in a three-level random-effects model. Pooled log response ratio **0.31** (0.25–0.37), $I^2$ 78 %, prediction interval 0.11–0.63. Factory-calibrated units read 43 % high, site-calibrated units 13 %. Site calibration was the strongest moderator ($\beta$ −0.24), then the log GSD of the size distribution ($\beta$ 0.15) and RH. **Where it is weak:** geometric-mean ratios pooled over very different aerosols and campaign lengths; the prediction interval says a single uncalibrated deployment can be off by 12–88 %. **Why it matters here:** no eligible study tested sub-$500 sensors against gravimetry. The moderators found for professional photometers — size distribution and humidity — are the terms low-cost calibration models already carry, and this gives them a pooled prior.

## What changed today

- **Electret filters fail on soot, not clogging.** Residential HVAC electret filters lose efficiency through soot-driven charge neutralisation at loadings far below any pressure-drop rise; modelled effective life during wildfire smoke falls from months to days.
- **Cooking leads primary $PM_1$ emission in London and Beijing.** Eddy-covariance AMS fluxes: cooking up to 87 % of early-evening primary $PM_1$ emission in London, 76 % in Beijing; traffic and combustion still lead in Delhi. Cooking is absent from national and global inventories.
- **Size-resolved number carries the summer signal.** Beijing oxylipin panel ($n = 123$): UFP associated with +12.9 to +97.7 % oxylipin change per IQR in summer, −7.7 to +18.8 % in winter.
- **Randomised roadside crossover.** 51 young adults: per 10 $\mu g\,m^{-3}$ $PM_{10}$, $FEF_{75}$ $-0.215\,L\,s^{-1}$; noise co-exposure appeared to amplify $PM_1/PM_{2.5}$ small-airway decrements (exploratory).
- **Cross-day resurfacing.** Three records indexed on 23 October reappeared in the 24 October Europe PMC and PubMed legs; they are carried here, the first issue to see them.
- **Held record released.** `10.1021/acs.est.6c07515` (NK-cell dysfunction from respirable mining dust), held unauthored from the 3 October issue, is now indexed in PubMed and carried here.

### Window continuity

The 22 October issue closed with `last_entry_date` at 22 October. This issue collects **23 October in full**, continuous with it. The scheduled runs for 23–29 October did not complete: a 30 October 00:21 UTC run harvested 23–28 October but stopped before screening. This issue was screened, authored and built on the 30 October 2026 14:20 UTC (09:20 CDT) recovery run, which backfills 23–29 October day by day; the W43 weekly (18–24 October) is built on the same run.

## Corpus analytics

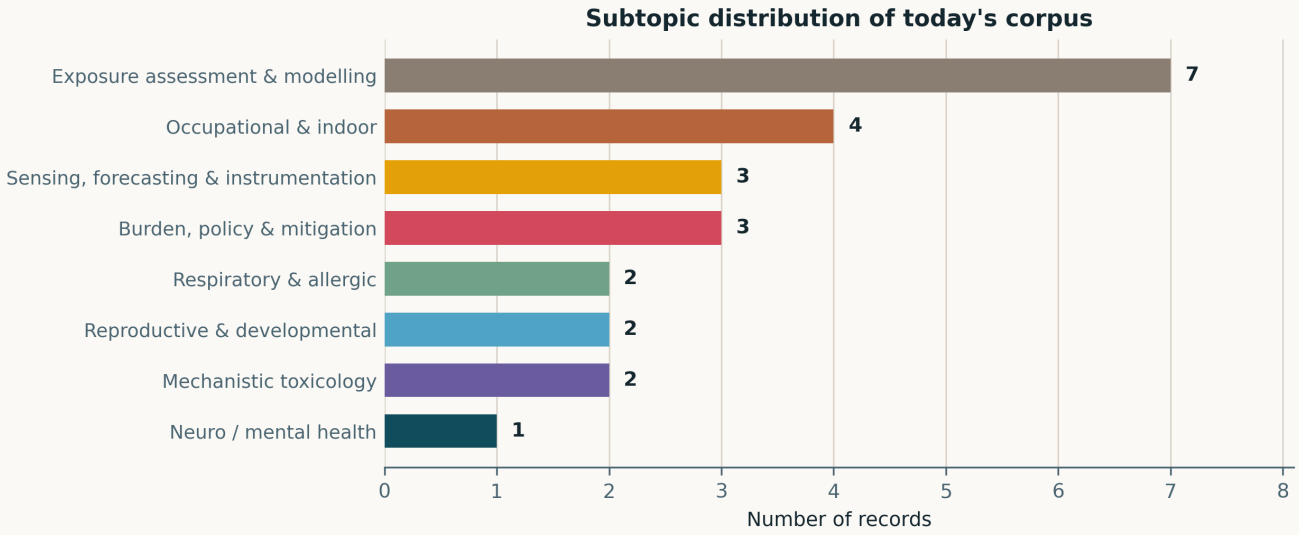



**Fig.** Subtopic assignment of the 24 in-scope records across eight bands. Exposure assessment leads with seven (flux source attribution, receptor modelling, brown carbon, NPF simulation and three metadata-only records); occupational/indoor four; sensing three. Cardiovascular and other clinical endpoints are empty.

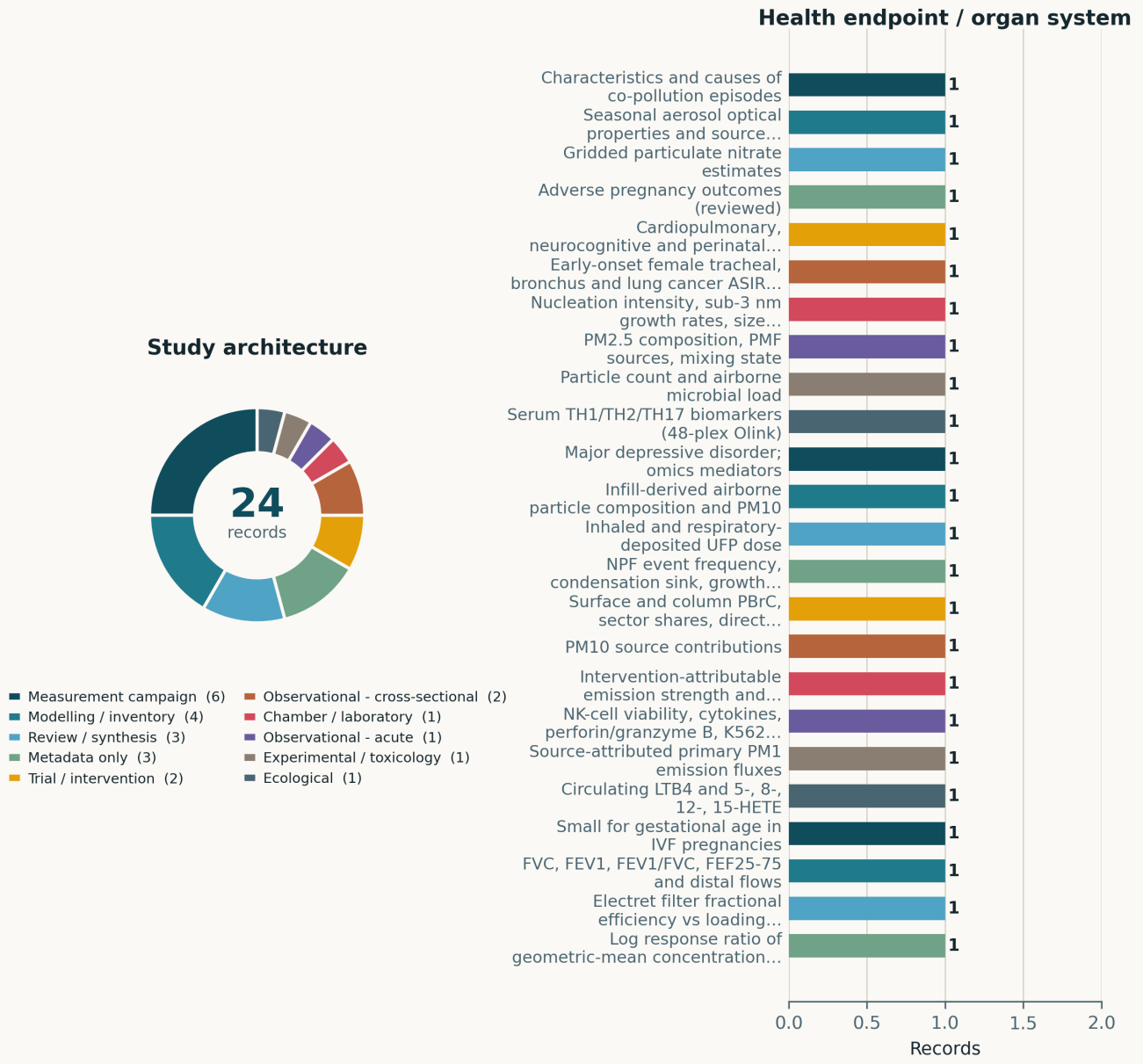



**Fig.** Left — architecture: two reviews, two field studies, one randomised crossover, one panel, one case-control, one meta-analysis, modelling, in-vitro and burden records; **metadata-only 3 (13 %)**. Right — endpoints run from SGA and depression to lung function, oxylipins, cytokines and instrument bias.

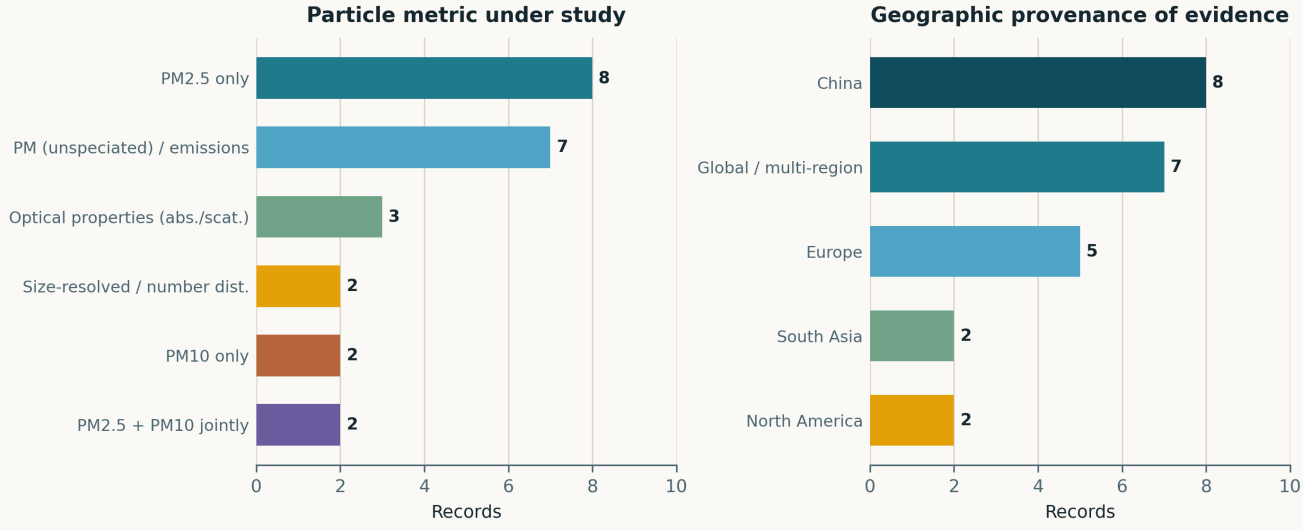



**Fig.** Left — particle metric: mass ($PM_{2.5}$/$PM_{10}$) dominates; number-based UFP metrics appear in four records (oxylipin panel, wood stoves, NPF field and model). Right — geography: China leads; Europe (Spain, Norway, UK) next; multi-region records are the meta-analysis, the three-megacity flux study, MR and GBD.

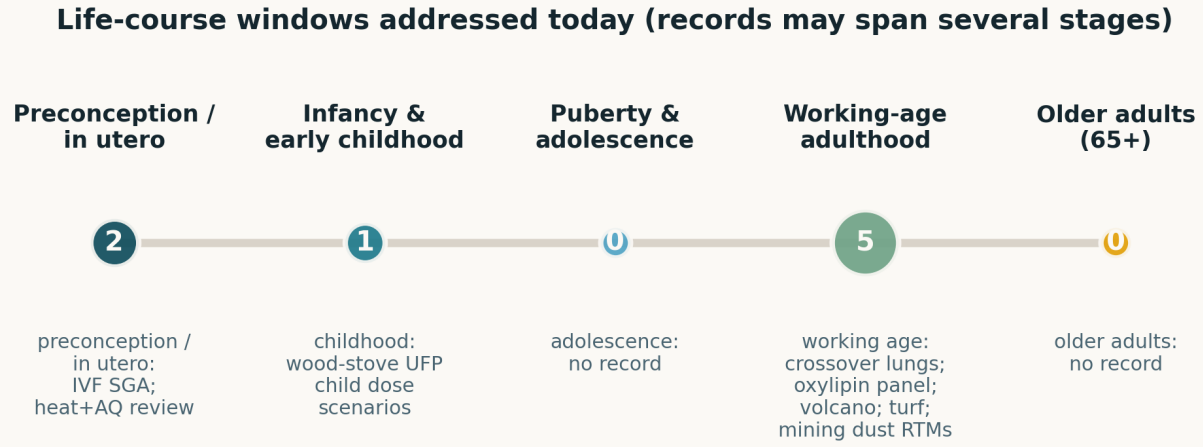



**Fig.** Life-course windows: two in-utero/pregnancy records (IVF SGA; heat and air-quality review), one childhood dose scenario (wood-stove UFP), five working-age records; no adolescent or older-adult record.

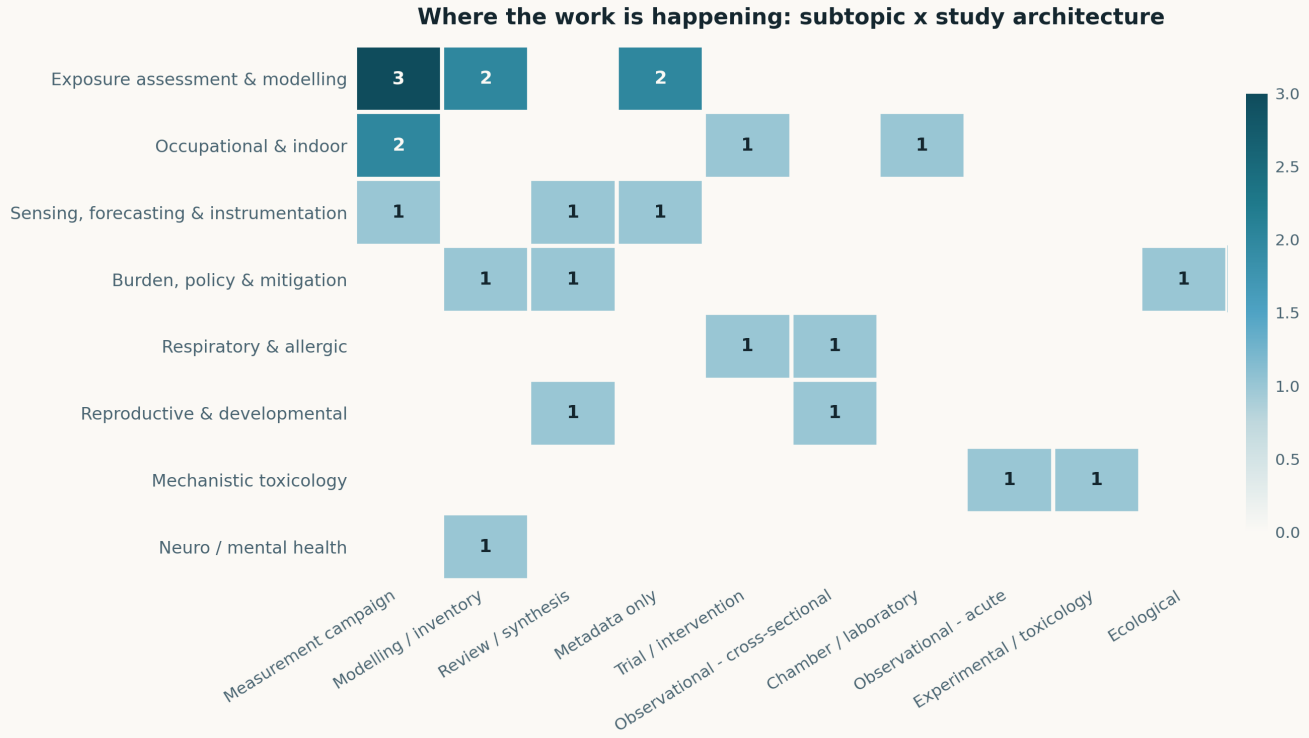



**Fig.** Subtopic × study architecture. Exposure assessment spreads across measurement, modelling and metadata-only cells; sensing holds the meta-analysis, a long-term field campaign and a metadata-only estimator; the health bands sit in the crossover, panel, case-control and MR cells.

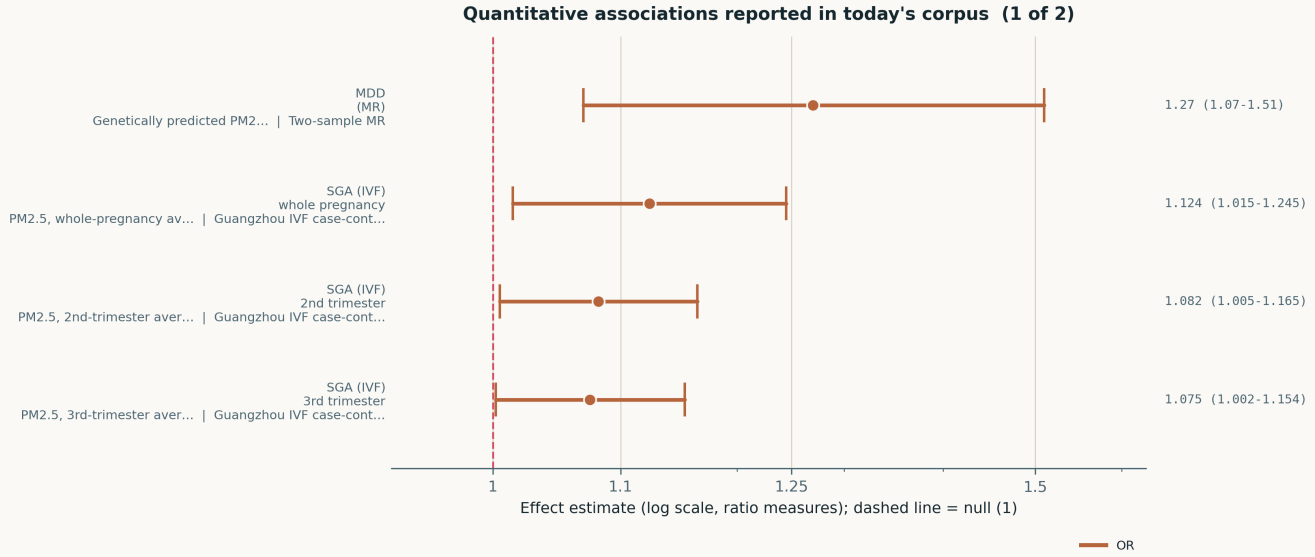



**Fig.** Ratio scale, four estimates from two studies: $PM_{2.5}$ and small for gestational age in IVF pregnancies (whole pregnancy, second and third trimester; Guangzhou matched case-control, increment not stated), and genetically predicted $PM_{2.5}$ and major depression (two-sample MR preprint, unit not stated). Not poolable.

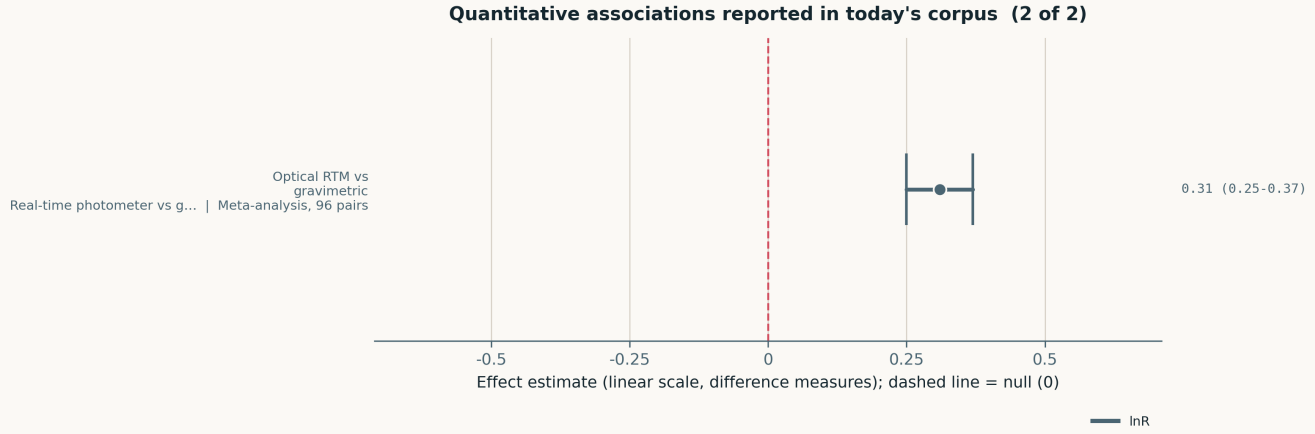



**Fig.** Difference scale: pooled log response ratio of optical real-time dust monitors vs gravimetric sampling (96 pairs, three-level random effects; $I^2$ 78 %). A value of 0.31 corresponds to a 36 % positive bias.

## Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

### Neuro / mental health — 1 record

**Wang multi-omics Mendelian randomization $PM_{2.5}$ depression 2026** Research Square (preprint)

*Multi-omics Mendelian Randomization Reveals the Causal Association and Mediating Mechanisms of $PM_{2.5}$ with Depression: A Systematic Study Integrating Transcriptomics, Metabolomics, and Radiomics*

LD-score regression found genetic correlation between 'PM2.5 exposure' and MDD (rg 0.334, P 1.9e-11); two-sample MR gave OR 1.27 (1.07-1.51) for MDD. Two-step MR identified five blood genes, six brain-region genes and two blood fatty-acid ratios as mediators; no mediation through CSF metabolites or imaging traits. Limitation: a GWAS of residential $PM_{2.5}$ indexes where people live, so its instruments capture SES, urbanicity and migration; the genetic correlation itself shows horizontal pleiotropy with MDD is likely, which violates the exclusion restriction. The OR has no exposure unit. Preprint. Relevance: low - not an exposure-measurement result.

doi: 10.21203/rs.3.rs-10994413/v1

## Respiratory & allergic — 2 records

**Meng traffic noise air pollution lung function randomised crossover 2026** Environ Res

*Independent and interactive effects of traffic-related noise and air pollution on lung function: A randomized crossover study in healthy young adults.*

Randomised crossover with four exposure conditions (noise and air pollution separated by intervention) in 51 healthy young adults beside a busy road at evening rush hour; pollutants and noise measured each session, mixed models adjusted for age, sex and BMI. Per 10 $\mu g\,m^{-3}$ $PM_{10}$, FEF75 fell 0.215 L/s; per 10 $\mu g\,m^{-3}$ $O_3$, 0.224 L/s. Noise independently reduced dynamic ventilatory indices, and exploratory interaction terms suggested noise amplified $PM_1$- and $PM_{2.5}$-associated FEV1 and small-airway decrements. No CIs in the abstract. Limitation: the four interventions are not described in the abstract and exposure conditions cannot be blinded, and pollutant contrasts within a single rush-hour session are small and collinear, so the interaction findings are hypothesis-generating. Relevance: medium - co-located acoustic and PM nodes are needed to separate the two stressors.

doi: 10.1016/j.envres.2026.125963

**Guerra-Ruiz cytokines volcanic emissions Cumbre Vieja ASHES 2026** Ecotoxicol Environ Saf

*Systemic cytokine signatures associated with volcanic emisions exposure and respiratory disease after the cumbre vieja eruption-ashes project.*

Cross-sectional ASHES sub-study after the 2021 Tajogaite eruption: 102 adults (66 general population, 13 occupationally exposed, 23 asthmatic). Workers had the longest exposure (median 10 h vs 6 h) and higher IL-2, IL-7, IL-17A and IL-17F; asthmatics showed broader TH1/TH2/TH17 activation, and IL-4 correlated with exposure time in asthmatics (r 0.79). Limitation: exposure is hours of presence, not measured PM or $SO_2$, and 13 workers is too few for a stable comparison; 48 markers tested without stated correction. Relevance: medium - the eruption had dense monitoring; linking those PM records to individuals would turn this into an exposure-response study.

doi: 10.1016/j.ecoenv.2026.120950

## Reproductive & developmental — 2 records

**Liu $PM_{2.5}$ components small for gestational age IVF Guangzhou 2026** BMC Public Health

*Maternal exposure to PM(2.5) and its components and risk of small for gestational age in women undergoing in vitro fertilization: a propensity score-matched case-control study in Guangzhou, China.*

Propensity-score-matched case-control within 1,459 IVF conceptions in Guangzhou: 166 SGA cases matched 1:1. Whole-pregnancy $PM_{2.5}$ OR 1.124 (1.015-1.245); second trimester 1.082 (1.005-1.165); third trimester 1.075 (1.002-1.154); the exposure increment is not given in the abstract. SO4, OM and BC were positively associated; mixture models ranked nitrate as the dominant driver, and no subgroup reached significance. Limitation: 332 matched women give lower CI bounds within 0.2-1.5% of the null across many component-by-window tests, with no multiplicity correction reported; the component estimates come from a modelled product whose component-specific error is unknown. Relevance: low-medium - component attribution here rests on modelled speciation that sensor networks cannot supply.

doi: 10.1186/s12889-026-28800-4

**Atkin air quality excess heat pregnancy outcomes review 2026** Nurs Clin North Am

*Impact of Poor Air Quality and Excess Heat on Pregnancy Outcomes.*

Narrative review on combined poor air quality and excess heat in pregnancy. Argues that physiological adaptations of pregnancy raise susceptibility, that structural determinants concentrate exposure in some populations, and that nurses should educate patients, promote adaptation and advocate for climate mitigation. No quantitative synthesis in the abstract. Limitation: non-systematic; heat-pollution interaction is asserted rather than quantified. Relevance: low - clinical guidance context only.

doi: 10.1016/j.cnur.2026.06.010

## Mechanistic toxicology — 2 records

**Wang seasonal PM size constituents oxylipin panel Beijing 2026** Environ Sci Technol

*Winter and Summer Differences in Oxylipin Responses to Ambient Particles: Roles of Sizes and Constituents*

Panel of 123 adults in Beijing with repeated visits in winter and summer; 27 $PM_{2.5}$ mass, size and constituent metrics against five lipoxygenase-derived oxylipins. Associations peaked at lag 0-1 d; across metrics, oxylipin increases ranged 3.3-114.2%, most robust for UFP, accumulation-mode number, OC and chloride. UFP showed an exposure-response only in summer: per IQR, +12.9 to +97.7% in summer vs -7.7 to +18.8% in winter; accumulation-mode and OC effects were larger in winter. No CIs in the abstract. Limitation: 27 metrics x 5 biomarkers x 2 seasons x several lags is a large test space, and the season contrast also changes time outdoors and ventilation. Relevance: high - size-resolved number, not mass, carries the summer signal, which mass-only optical sensors cannot report.

doi: 10.1021/acs.est.6c08647

**Sun respirable mining dust natural killer cell dysfunction 2026** Environ Sci Technol

*Respirable Particulate Matter Induces Natural Killer Cell Dysfunction via Oxidative Stress-Mediated Lysosomal and Mitochondrial Injury: Implications for Mineral-Rich Mining Dust Exposure.*

In-vitro exposure of NK cells to respirable mineral dusts with a K562 target-cell challenge. Injury was dose- and composition-dependent: apoptosis/necrosis, shifted cytokines, altered perforin/granzyme B function and impaired killing. Silica- and silicate-rich dusts were worst; iron-bearing and carbonaceous dusts produced distinct oxidative patterns. Mitochondrial depolarisation, lower mitochondrial mass and lysosomal deacidification converged across dusts; ascorbic acid partially restored organelle function and IFN-gamma. Doses and their relation to inhaled lung burden are not given in the abstract. Limitation: an isolated cell line in submerged culture, no lung-burden dosimetry, so the proposed adverse outcome pathway to lung cancer is inferential. Relevance: low - composition (silica fraction) matters, which mass sensors cannot resolve. Note: this DOI was held from the 3 Oct issue as unauthored and is now carried here.

doi: 10.1021/acs.est.6c07515

## Exposure assessment & modelling — 7 records

**Langford cooking dominant primary $PM_1$ eddy covariance London Beijing Delhi 2026** npj Clim Atmos Sci

*Traffic and combustion emission controls leave cooking as a dominant local source of primary $PM_1$ in megacities*

Eddy-covariance fluxes with aerosol mass spectrometry and a new source-attribution method in three megacity neighbourhoods. In Delhi, traffic and combustion dominate primary $PM_1$ emission. In London and Beijing, cooking is the largest local primary source at mealtimes: up to 83% of midday and 87% of early-evening emission in London, 61% and 76% in Beijing. Cooking is absent from national, European and global inventories, though the London Atmospheric Emissions Inventory's estimate matched. Primary emission sources differ from those controlling ambient $PM_1$ mass, which is secondary-dominated. Limitation: one flux footprint per city and non-refractory $PM_1$ only (no BC or dust). Relevance: high - source-resolved flux benchmarks for inventory checks; cooking plumes are a known optical-sensor artefact at street level.

doi: 10.1038/s41612-026-01583-6

**Perez-Vizcaino $PM_{10}$ sources Riotinto mining multi-time-resolution PMF 2026** J Hazard Mater

*Characterization of PM(10) sources in the Riotinto copper mining district (SW Spain) using multi-time-resolution receptor modelling.*

Year-long (July 2024-July 2025) campaign combining an hourly XRF metals monitor and an Aethalometer with daily high-volume quartz-filter chemistry, merged in multi-time-resolution PMF. Seven sources: waste rock 4.9, regional + traffic 3.6, open pit 1.6, combustion 1.2, aged marine 1.2, dust 1.1 and copper concentrate 0.7 $\mu g\,m^{-3}$. Daily species identified traffic tracers and SOA that the hourly subset alone missed. Arsenic was the main geochemical anomaly. Limitation: one receptor site; multi-resolution PMF weights the daily data heavily and factor uncertainty (bootstrap/DISP) is not reported in the abstract. Relevance: medium - hourly XRF + BC is the reference tier that low-cost networks around mines would need to be anchored to for metal-rich dust.

doi: 10.1016/j.jhazmat.2026.143869

**Chen primary brown carbon China 2005-2020 trends radiative effect 2026** Atmos Chem Phys

*Long-term evolution and effects of primary brown carbon aerosol in China*

CTM simulation of primary BrC across China, 2005-2020, with an internal-mixing optical model. Mean surface PBrC 0.81 ugC/m3; residential, industrial and agricultural combustion contributed 57%, 22% and 18%, together 91% of column PBrC in 2010. PBrC fell 20.8% over the period vs 8.1% for $PM_{2.5}$; direct radiative effect fell from +0.032 to +0.023 W/m2 (-25.7%), 84.2% anthropogenic. Limitation: BrC emission factors and absorption Angstrom exponents are poorly constrained, and evaluation against measured absorption is not described in the abstract. Relevance: medium - multi-wavelength absorption (BrC) channels on sensor nodes could test the residential-combustion dominance.

doi: 10.5194/acp-26-14867-2026

**Liu $PM_{2.5}$ sources mixing state Chengdu Kangding altitude 2026** Environ Monit Assess

*Altitudinal and seasonal differences in PM(2.5) sources, secondary formation, and particle mixing states in the basin-plateau transition zone of Southwest China.*

Summer and winter $PM_{2.5}$ sampling at Chengdu (500 m) and Kangding (2,500 m). Winter $PM_{2.5}$ 79.4 vs 15.2 $\mu g\,m^{-3}$, summer 29.1 vs 8.1. Inorganic ions dominated in Chengdu, carbonaceous species in Kangding; sulfate in summer gave way to nitrate in winter. PMF: Chengdu combustion 26.0%, traffic 32.4%, secondary 35.3%; Kangding combustion 47.9%. Single-particle data showed more SIA particles in Chengdu (29.0% vs 17.6%) and more organic-SIA internal mixing in Kangding. Limitation: two seasons, two sites; the altitude contrast is confounded with city size and heating fuel. Relevance: low - descriptive regional chemistry.

doi: 10.1007/s10661-026-16102-6

**Ghosh meter-scale heterogeneity new particle formation 3D model 2026** Environ Sci Technol

*How Microscale Flow and Vapor Transport Shape New Particle Formation and Growth.*

A 10 m resolution 3D multiphysics aerosol model (3D-ICAM) compared with an equivalent box model across emission geometries. Under weak winds (<1 m/s), it reproduced nucleation hotspots, intermittent multi-episode nucleation and persistent bimodal size distributions that the homogeneous case lacks. Resolving spatial structure raised predicted sub-3 nm growth rates up to ~20-fold, bringing them into the observed range, because transport-generated microenvironments relax the growth-scavenging competition. Limitation: idealised geometries; agreement with observations is at the level of growth-rate ranges, not site-specific validation. Relevance: medium - implies point UFP measurements in streets sample sub-grid hotspots; dense number-sensor networks are the natural test.

doi: 10.1021/acs.est.6c10206

**Kaskaoutis in situ aerosol optical properties eastern Mediterranean 2026** Atmos Environ

*In situ aerosol optical properties in the background eastern Mediterranean: seasonal variability, regional transport, and source attribution*

METADATA ONLY. Title and authors only (Kaskaoutis, Baalbaki, Bimenyimana et al.). No abstract was retrievable at harvest. The title describes in situ scattering/absorption measurements at a background eastern Mediterranean site, with seasonal variability, regional transport and source attribution. Site, instruments and values cannot be stated. Limitation: unverifiable beyond the title. Relevance: low-medium - background optical properties set the mass-scattering efficiency that optical PM sensors assume.

doi: 10.1016/j.atmosenv.2026.122436

**Zhao $PM_{2.5}$-$O_3$ co-pollution episodes BTH YRD 2026** Atmos Pollut Res

*Episode analysis of the characteristics and causes of $PM_{2.5}$–$O_3$ co-pollution in Beijing-Tianjin-Hebei and Yangtze River Delta, China*

METADATA ONLY. Title and authors only (Zhao, Xu, Gao, Zhang et al.). No abstract was retrievable at harvest. The title describes an episode analysis of simultaneous $PM_{2.5}$ and $O_3$ exceedance in BTH and the YRD, including causes. Data sources, period and attribution method cannot be stated. Limitation: unverifiable beyond the title. Relevance: low-medium - co-pollution episodes are where PM-only alerts from sensor networks understate risk.

doi: 10.1016/j.apr.2026.103248

## Sensing, forecasting & instrumentation 3 records

**Sim real-time dust monitor bias gravimetric meta-analysis 2026** Ann Work Expo Health

*Systematic bias in real-time dust monitoring: a meta-analysis of measurement discrepancy and its implications for occupational exposure assessment.*

PRISMA meta-analysis of 112 occupational studies; 96 paired RTM-gravimetric comparisons pooled in a three-level random-effects model. Professional photometers read 36% high on average: pooled lnR 0.31 (0.25-0.37), I2 78.4%, prediction interval 0.11-0.63. Factory-calibrated units read 43% high, site-calibrated units 13% high; site-specific calibration was the strongest moderator (beta -0.24, P<0.001), followed by log GSD of the size distribution (beta 0.15) and RH (beta 0.002 per %). No eligible study compared sub-$500 sensors against gravimetry. Limitation: geometric-mean ratios from heterogeneous aerosols and campaign lengths; the prediction interval never crosses zero but its width means a single uncalibrated deployment can be off by 12-88%. Relevance: high - the size-distribution and RH moderators are the same terms LCS calibration models need, now with a pooled prior.

doi: 10.1093/annweh/wxag089

**Srivastav ion particle size distribution NPF Indo-Gangetic Plain 2026** Atmos Chem Phys

*Ion and particle size distribution measurements in the Indo-Gangetic Plain: seasonal insights and challenges in new particle formation event characterization*

Ion and particle spectrometer measurements at a suburban IGP site, May-December 2023. NPF occurred on 30/35 valid days (86%) in May-June but 20/73 (28%) in winter/post-monsoon. Morning median condensation sink was lower on event days (0.0095 vs 0.0292 s-1). Growth rates rose with size, 14.7 nm/h (3-7 nm) to 19.0 nm/h (7-20 nm). Pollution plumes intermittently masked or distorted events, so both visual and automated classification and growth-rate fitting were unreliable. Limitation: no precursor gas (H2SO4, amines, HOMs) measurements, so mechanism is unconstrained; eight months at one site. Relevance: medium - in high-CS settings the UFP number seen by condensation or diffusion-charger sensors is plume- not nucleation-driven.

doi: 10.5194/acp-26-14917-2026

**Cai self-supervised stacking particulate nitrate sparse observations 2026** Atmos Environ

*A Self-supervised Stacking Framework for Estimating Particulate Nitrate in Two Chinese Urban Agglomerations under Sparse Ground Observations*

METADATA ONLY. Title and authors only (Cai, He, Li, Liu et al.). No abstract was retrievable from Crossref, Europe PMC or Semantic Scholar at harvest. The title describes a self-supervised stacking ensemble that estimates particulate nitrate in two Chinese urban agglomerations from sparse ground observations. Predictors, validation design and skill cannot be stated. Limitation: unverifiable beyond the title; with sparse speciation sites, random cross-validation would overstate skill (see the 22 Oct blocked-CV record). Relevance: medium - component estimation under sparse reference sites is the same problem as filling between LCS anchors.

doi: 10.1016/j.atmosenv.2026.122433

## Occupational & indoor 4 records

**Wang electret HVAC filter charge neutralisation wildfire smoke 2026** Environ Sci Technol

*Charge-Driven Degradation of Residential Electret HVAC Filters under Wildfire Smoke Exposure.*

Laboratory flaming and smouldering wood smoke, field sampling at prescribed burns and controlled soot challenges on residential electret HVAC filters. Filtration efficiency collapsed at particle loadings far below those that raise pressure drop; the driver was soot-rich particles neutralising electret charge irreversibly, not mechanical clogging or organic-aerosol deposition. Built into a dynamic indoor model driven by EPA monitor data for a regional wildfire event, effective filter life fell from months to days at moderate outdoor $PM_{2.5}$, with protection varying by building and ventilation. No efficiency numbers are given in the abstract. Limitation: few filter products and smoke types; charge loss in the field depends on soot fraction, which varies by fire stage. Relevance: high - indoor/outdoor LCS pairs can detect the efficiency collapse that pressure-based replacement rules miss.

doi: 10.1021/acs.est.6c03046

**Kuye ultrafine particle respiratory deposition dose wood stoves UK homes 2026** Atmosphere

*Respiratory Deposition Dose of Ultrafine Particles from Improved Wood Stoves and Open Fireplaces*

Exploratory measurements in four UK homes (Ecodesign stove, multifuel Ecodesign stove x2, open fireplace), 11-12 winter days each, 1-min PNC combined with age/sex/activity ventilation rates and the ICRP model. Daily inhaled dose 0.45-16.0 x10^9 particles/kg/day sitting and 2.37-73.2 x10^9 at moderate activity (4.5-5.3-fold higher). Child scenarios had 2.7-3.8x the adult body-mass-normalised dose; household mean deposition rate 1.50-5.99 x10^8 particles/min. Limitation: n=4 homes that differ simultaneously in appliance, fuel, room, ventilation and behaviour, so stove-type effects cannot be identified; doses are modelled from concentration, not measured. Relevance: medium - residential UFP is invisible to mass sensors; indoor PNC nodes are the relevant instrument.

doi: 10.3390/atmos17111039

**Stange SEM-Raman inhalable particles indoor artificial turf infill 2026** Research Square (preprint)

*Correlative SEM–Raman Microscopy of Inhalable Particles Sampled during Indoor Football on Artificial Turf with Crumb Rubber and Olive-Stone Infills*

Air sampling during matches in two Norwegian indoor halls, one with crumb-rubber infill and one with olive-stone/quartz infill, analysed by correlative SEM, AI particle classification and micro-Raman. Both infills released airborne particles. Olive-stone particles appeared indoors only; tyre rubber appeared indoors and outdoors, implicating traffic as well as infill. $PM_{10}$ was 58.4 $\mu g\,m^{-3}$ in the olive-stone hall vs 6.0 $\mu g\,m^{-3}$ with crumb rubber, and fungal spores were found with the organic infill. Limitation: one hall per infill, so the tenfold $PM_{10}$ difference is confounded by hall ventilation, occupancy and cleaning. Preprint. Relevance: low-medium - substituting a 'sustainable' material changed composition and raised mass.

doi: 10.21203/rs.3.rs-11040166/v1

**Mohammed mobile HEPA-UV air decontamination operating theatre 2026** Infect Prev Pract

*Evaluation of a mobile surgical air decontamination device in reducing airborne microbial load in the operating theatre of a tertiary care hospital.*

Before-after evaluation of a mobile HEPA + UV decontamination unit in one operating theatre over 15 days (July 2025). Mean total particle count fell from 6,836 to 4,072 particles/m3 (-40.4%), significant in the 0.3 and 0.5 um channels; CFU fell from 34.5 to 15.2 CFU/m3 (P 0.003). Limitation: one theatre, short before-after design with no control period or blinding to activity level, and no clinical outcome (surgical-site infection). Relevance: low-medium - shows optical counters as the performance metric for portable cleaners in clinical rooms.

doi: 10.1016/j.infpip.2026.100583

## Burden, policy & mitigation 3 records

**Liang emergency haze controls causal ML synthetic control Shijiazhuang 2026** Environ Sci Technol

*Evaluating Emergency Emission Controls During an Unprecedented Regional Haze Event Using Causal Machine Learning*

Three-step causal evaluation (meteorological normalisation, augmented synthetic control, ML prediction) of Shijiazhuang's 45-day 'Sharp Sword' campaign in 2016. Local emission strength fell 4.3% ($PM_{2.5}$), 4.8% ($PM_{10}$), 15.8% ($SO_2$), 5.8% ($NO_2$), 15.3% (CO); concentrations fell 42.1 $\mu g\,m^{-3}$ (14.8%) for $PM_{2.5}$ and 47.4 $\mu g\,m^{-3}$ (12.4%) for $PM_{10}$. Independent PMF showed vehicle, steelmaking and coal contributions 4.1%, 11.0% and 21.7% lower than in 2017. $PM_{2.5}$ still averaged 243.2 $\mu g\,m^{-3}$ during the campaign. Limitation: donor-city synthetic controls share the regional haze and possibly spillover controls, biasing the effect toward zero; no uncertainty intervals in the abstract. Relevance: medium - a transferable counterfactual framework for evaluating interventions from network data.

doi: 10.1021/acs.est.6c12991

**Zhang early-onset female lung cancer burden GBD 2021 2026** Asia Pac J Clin Oncol

*Global Burden and Trends of Early-Onset Tracheal, Bronchus, and Lung Cancer Among Women From 1990 to 2021.*

GBD 2021 secondary analysis of early-onset (under 50) female TBL cancer, 1990-2021. Global rates declined (all EAPCs <0), but low-middle SDI regions rose (DALY EAPC 0.66) and South Asia fastest (DALY EAPC 1.42). Ambient PM was the fastest-growing attributable risk (attributable DALY rate EAPC 1.22, ASDR 1.27). Limitation: GBD attributable burden is model output that inherits the exposure surface and IER curves; in young women with few cases, estimates in low-SDI regions rest largely on modelled covariates rather than registries. Relevance: low - burden, not measurement.

doi: 10.1111/ajco.70212

**Cantu air quality across the lifespan nursing response review 2026** Nurs Clin North Am

*Breathing Unequal Air: Global and US Air Quality Across the Lifespan and the Nursing Response.*

Narrative review for nurses of global and US air quality trends and health effects across the lifespan, citing ~8 million deaths/year attributable to air pollution, with $PM_{2.5}$, $O_3$ and $NO_2$ as the principal pollutants. Covers cardiopulmonary, neurocognitive and perinatal outcomes, nursing practice and policy, and argues for integrating air-quality awareness into nursing curricula. No new data. Limitation: non-systematic; no synthesis of effect sizes. Relevance: low - possible audience for patient-facing use of public sensor maps.

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# Corpus register

| First author | Focus | Journal | Design | Metric | Region |
|---|---|---|---|---|---|
| **Neuro / mental health** | | | | | |
| Wang multi-omics Mendelian randomization $PM_{2.5}$ depression | Multi-omics Mendelian Randomization Reveals the Causal Association and Mediating Mechanisms of $PM_{2.5}$ with Depression: A Systematic Study Integrating Transcriptomics, Metabolomics, and Radiomics | *Research Square (preprint)* | Two-sample Mendelian randomisation | Genetically predicted $PM_{2.5}$ exposure (GWAS instruments) | Global / multi-region (GWAS summary data) |
| **Reproductive & developmental** | | | | | |
| Atkin air quality excess heat pregnancy outcomes review | Impact of Poor Air Quality and Excess Heat on Pregnancy Outcomes. | *Nurs Clin North Am* | Narrative review | Ambient air pollution and heat (reviewed) | Not stated in record |
| Liu $PM_{2.5}$ components small for gestational age IVF Guangzhou | Maternal exposure to PM(2.5) and its components and risk of small for gestational age in women undergoing in vitro fertilization: a propensity score-matched case-control study in Guangzhou, China. | *BMC Public Health* | Case-control study | Trimester-averaged $PM_{2.5}$ and SO4, NO3, NH4, OM, BC (gridded estimates at residence) | China (Guangzhou) |
| **Respiratory & allergic** | | | | | |
| Guerra-Ruiz cytokines volcanic emissions Cumbre Vieja ASHES | Systemic cytokine signatures associated with volcanic emisions exposure and respiratory disease after the cumbre vieja eruption-ashes project. | *Ecotoxicol Environ Saf* | Observational - cross-sectional | Volcanic gas and particulate exposure (hours of exposure, group membership) | Spain (La Palma) |
| Meng traffic noise air pollution lung function randomised crossover | Independent and interactive effects of traffic-related noise and air pollution on lung function: A randomized crossover study in healthy young adults. | *Environ Res* | Randomised crossover trial | Roadside PM1, $PM_{2.5}$, $PM_{10}$, O3, NO2 and noise during evening rush hour | China (urban roadside) |
| **Mechanistic toxicology** | | | | | |
| Sun respirable mining dust natural killer cell dysfunction | Respirable Particulate Matter Induces Natural Killer Cell Dysfunction via Oxidative Stress-Mediated Lysosomal and Mitochondrial Injury: Implications for Mineral-Rich Mining Dust Exposure. | *Environ Sci Technol* | In vitro | Compositionally distinct respirable PM (silica/silicate, iron-bearing, carbonaceous mine dusts) | Not stated in record |
| Wang seasonal PM size constituents oxylipin panel Beijing | Winter and Summer Differences in Oxylipin Responses to Ambient Particles: Roles of Sizes and Constituents | *Environ Sci Technol* | Panel study | 27 $PM_{2.5}$ metrics: mass, UFP and accumulation-mode number, OC, chloride and other constituents | China (Beijing) |
| **Occupational & indoor** | | | | | |
| Kuye ultrafine particle respiratory deposition dose wood stoves UK homes | Respiratory Deposition Dose of Ultrafine Particles from Improved Wood Stoves and Open Fireplaces | *Atmosphere* | Field measurement | Indoor particle number concentration (1-min) with ICRP deposition model | United Kingdom (four homes) |
| Mohammed mobile HEPA-UV air decontamination operating theatre | Evaluation of a mobile surgical air decontamination device in reducing airborne microbial load in the operating theatre of a tertiary care hospital. | *Infect Prev Pract* | Intervention field trial (non-randomised) | Optical particle counts (0.3, 0.5 um and larger) and culturable CFU | India (New Delhi) |
| Stange SEM-Raman inhalable particles indoor artificial turf infill | Correlative SEM–Raman Microscopy of Inhalable Particles Sampled during Indoor Football on Artificial Turf with Crumb Rubber and Olive-Stone Infills | *Research Square (preprint)* | Field study | $PM_{10}$ mass; single-particle SEM + AI classification + micro-Raman of airborne dust | Norway (two indoor sports halls) |
| Wang electret HVAC filter charge neutralisation wildfire smoke | Charge-Driven Degradation of Residential Electret HVAC Filters under Wildfire Smoke Exposure. | *Environ Sci Technol* | Materials / filtration | Flaming/smouldering wood smoke, prescribed-burn smoke and soot challenges; EPA monitor $PM_{2.5}$ for model inputs | United States (laboratory, prescribed burns, regional wildfire scenario) |

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| **Sensing, forecasting & instrumentation** | | | | | |
| Cai self-supervised stacking particulate nitrate sparse observations | A Self-supervised Stacking Framework for Estimating Particulate Nitrate in Two Chinese Urban Agglomerations under Sparse Ground Observations | *Atmos Environ* | Metadata only (no abstract) | Particulate nitrate (estimated from sparse ground observations) | China (two urban agglomerations) |
| Sim real-time dust monitor bias gravimetric meta-analysis | Systematic bias in real-time dust monitoring: a meta-analysis of measurement discrepancy and its implications for occupational exposure assessment. | *Ann Work Expo Health* | Systematic review + random-effects meta-analysis | Optical real-time dust monitors (DustTrak, SidePak, pDR) vs gravimetric samplers | Global / multi-region (occupational literature 2010-2025) |
| Srivastav ion particle size distribution NPF Indo-Gangetic Plain | Ion and particle size distribution measurements in the Indo-Gangetic Plain: seasonal insights and challenges in new particle formation event characterization | *Atmos Chem Phys* | Long-term field measurement campaign | Air-ion (0.8-42 nm) and particle (2.5-42 nm) size distributions | India (Sonipat, Haryana) |
| **Exposure assessment & modelling** | | | | | |
| Chen primary brown carbon China 2005-2020 trends radiative effect | Long-term evolution and effects of primary brown carbon aerosol in China | *Atmos Chem Phys* | Chemical transport modelling | Modelled primary brown carbon (internal-mixing optical model), $PM_{2.5}$ | China |
| Ghosh meter-scale heterogeneity new particle formation 3D model | How Microscale Flow and Vapor Transport Shape New Particle Formation and Growth. | *Environ Sci Technol* | Theory / simulation | Simulated nucleation-mode particle number and size distributions (3D-ICAM, 10 m resolution) | Not stated in record |
| Kaskaoutis in situ aerosol optical properties eastern Mediterranean | In situ aerosol optical properties in the background eastern Mediterranean: seasonal variability, regional transport, and source attribution | *Atmos Environ* | Metadata only (no abstract) | In situ aerosol scattering and absorption | Not stated in record (eastern Mediterranean) |
| Langford cooking dominant primary PM1 eddy covariance London Beijing Delhi | Traffic and combustion emission controls leave cooking as a dominant local source of primary PM1 in megacities | *npj Clim Atmos Sci* | Eddy covariance flux measurement, method development | Chemically resolved non-refractory PM1 fluxes (aerosol mass spectrometry + eddy covariance) | Global / multi-region (London, Beijing, Delhi) |
| Liu $PM_{2.5}$ sources mixing state Chengdu Kangding altitude | Altitudinal and seasonal differences in PM(2.5) sources, secondary formation, and particle mixing states in the basin-plateau transition zone of Southwest China. | *Environ Monit Assess* | Source apportionment | Filter $PM_{2.5}$ chemistry (ions, carbon) and single-particle mass spectrometry | China (Chengdu and Kangding, Sichuan) |
| Perez-Vizcaino $PM_{10}$ sources Riotinto mining multi-time-resolution PMF | Characterization of PM(10) sources in the Riotinto copper mining district (SW Spain) using multi-time-resolution receptor modelling. | *J Hazard Mater* | Intensive field campaign, PMF source apportionment | Hourly XRF elements and Aethalometer BC + daily filter $PM_{10}$ chemistry | Spain (Riotinto mining district) |
| Zhao $PM_{2.5}$-O3 co-pollution episodes BTH YRD | Episode analysis of the characteristics and causes of $PM_{2.5}$–O3 co-pollution in Beijing-Tianjin-Hebei and Yangtze River Delta, China | *Atmos Pollut Res* | Metadata only (no abstract) | $PM_{2.5}$ and O3 co-pollution episodes (monitoring data) | China (Beijing-Tianjin-Hebei and Yangtze River Delta) |
| **Burden, policy & mitigation** | | | | | |
| Cantu air quality across the lifespan nursing response review | Breathing Unequal Air: Global and US Air Quality Across the Lifespan and the Nursing Response. | *Nurs Clin North Am* | Narrative review | $PM_{2.5}$, O3, NO2 (reviewed) | Global / multi-region (with United States focus) |
| Liang emergency haze controls causal ML synthetic control Shijiazhuang | Evaluating Emergency Emission Controls During an Unprecedented Regional Haze Event Using Causal Machine Learning | *Environ Sci Technol* | Augmented synthetic control + weather normalisation | Monitor $PM_{2.5}$, $PM_{10}$, SO2, NO2, CO; $PM_{2.5}$ source apportionment | China (Shijiazhuang) |
| Zhang early-onset female lung cancer burden GBD 2021 | Global Burden and Trends of Early-Onset Tracheal, Bronchus, and Lung Cancer Among Women From 1990 to 2021. | *Asia Pac J Clin Oncol* | Burden estimation (GBD) | Ambient particulate matter pollution (GBD risk factor) | Global / multi-region (GBD 2021) |

### Provenance and method

Entry window **23 October 2026**, single day (Friday). Harvester legs: **PubMed** health **67**, sensing **10**; **Europe PMC 40**; **Crossref by ISSN 70**; **arXiv** 150 kB, 1 in-window entry (rejected); **OpenAlex** HTTP 429. 50 cross-source duplicates removed; 128 fresh; PubMed connector (EDAT 23 Oct) 64 PMIDs, 20 appended, **148 screened**. Abstract retry recovered none of the three Elsevier metadata-only records. **24 admitted**, **124 rejected** with logged reasons. **DOI gate** 1 warn (nursing review title overlap). **Consensus** sweep and **trial watch**: run once for the 23–29 October backfill — see the 29 October issue and the W43 weekly. **3 carried on metadata alone**. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.